

Project Title:

Healthcare Infrastructure Security & Threat Response

Organization: MediCore Health Systems

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Role: Security Operations Center (SOC) Analyst

Tools Used:

Draw.io, Oracle VirtualBox, pfSense, Wazuh, Kali Linux,
Ubuntu Server, Wireshark.

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Executive Summary

Medicore healthcare systems is a healthcare organisation that recently expanded its infrastructure by launching a DMZ-hosted patient portal for telehealth services and online access to electronic health records (EHRs). However, this expansion has exposed critical security gaps. I investigated a specific incident where unusual traffic originated from internal workstations and moved towards our Electronic Health Records (EHRs) database and clinical systems. The central part is lack of effective network segmentation, which creates a risk of data breaches, HIPAA violations, and potential disruption to patient care.

To address this, I have designed and implemented segmented network architecture using Draw.io tool and pfSense to enforce firewall policies, isolating the clinical systems, administrative devices, medical devices, and the DMZ- hosted patient portal. Security monitoring was established using Wazuh, while Wireshark provided the network traffic analysis. Secondly, Kali Linux was also used to stimulate common cyberattacks, and the Ubuntu server hosted the security monitoring infrastructure.

Furthermore, I stimulated the attack path using the Oracle VirtualBox. The stimulated attacks included network reconnaissance, unauthorised access attempts, and lateral movement between network segments. This assessment demonstrated that effective network segmentation combined with continuous monitoring will significantly improve visibility, reduce the surface of attack, and limit the potential impact of cyber incidents.

In conclusion, the assessment achieved its objectives by improving Medicore Health Systems network security and incident detection abilities. Key recommendations include maintaining robust network segmentation, revising firewall rules on a regular basis, conducting vulnerability assessments, consistently monitoring security events, and providing workers with continuing cybersecurity awareness training. These enhancements promote regulatory compliance, safeguard sensitive patient information, and help secure and dependable delivery of healthcare services.

Introduction

MediCore Health Systems is a healthcare entity that oversees sensitive clinical data, encompassing patient records, appointment information, medical histories, and various types of protected health information (PHI). With the organisation's growth in digital capabilities via a DMZ-hosted patient portal, the necessity for enhanced cybersecurity measures became essential to maintain confidentiality, integrity, and availability of healthcare services. The growth of network infrastructure brought about new security issues, especially concerning unauthorised access, inadequate network segmentation, and restricted visibility into internal operations. In an internal evaluation, atypical communication behaviours were discovered from employee workstations trying to access essential clinical resources, such as systems assisting the EHR environment. These results highlighted concerns about possible lateral movement, exposure of confidential patient data, and the chance of security events impacting healthcare operations. Insufficient access controls and restricted monitoring abilities heightened the organisation's vulnerability to cyber threats and compliance issues. This SOC capstone project was designed to emulate the functions of a security operations team tasked with detecting, analysing, and mitigating cybersecurity threats in a healthcare setting. The initiative seeks to show how security experts can enhance organisational resilience by implementing effective network architecture, segmentation tactics, firewall enforcement, threat monitoring, and incident analysis. The simulation emphasises creating a secure network setting, applying safeguarding measures, examining questionable behaviour, and assessing the efficacy of security monitoring tools. Utilising tools like Draw.io, Oracle VirtualBox, pfSense, Wazuh, Kali Linux, Ubuntu Server, and Wireshark, the project offers hands-on experience in recognising vulnerabilities, detecting simulated attacks, and suggesting enhancements. The results of this study emphasise the significance of proactive security management in safeguarding healthcare data and assisting compliance with regulations.

Methodology

This project adopted a practical, scenario-based methodology to investigate and strengthen the security posture of MediCore Health Systems. A simulated healthcare environment was developed to replicate the organisation's network infrastructure, allowing security controls to be implemented and evaluated in a safe and controlled setting. The methodology followed a structured process comprising network design, implementation, monitoring, attack simulation, and security assessment.

The initial phase involved examining the organisation's current infrastructure and identifying vulnerabilities that might make sensitive clinical systems exposed to cyber threats. Considering the business needs and recognised risks, an enterprise network structure was created with Draw.io to illustrate the logical division of clinical systems, administrative machines, medical devices, and the publicly accessible patient portal situated within a demilitarised zone (DMZ).

To validate the proposed design, a virtual environment was created using Oracle VirtualBox. Individual virtual machines were configured to represent the various network segments, including workstations, servers, monitoring systems, and attacker machines. pfSense was deployed as the perimeter firewall and router to enforce network segmentation through Virtual Local Area Networks (VLANs), firewall rules, and access control policies. These measures were intended to minimise unnecessary communication between network zones and reduce opportunities for lateral movement.

A centralised security monitoring system was created with Wazuh, set up on an Ubuntu Server, to gather security events, observe system activity, and trigger alerts for unusual behaviour. Simultaneously, Wireshark was utilised to monitor and examine network traffic, offering further insight into communications throughout the environment and aiding in the verification of firewall settings. Kali Linux was utilised to assess the efficacy of the deployed controls by simulating real-world attack scenarios, such as network reconnaissance, port scanning, unauthorised access attempts, and lateral movement across network segments. These activities allowed for the evaluation of Wazuh's detection abilities while verifying if the firewall rules effectively limited malicious traffic.

The final phase involved examining the evidence gathered during the investigation. Security alerts, firewall records, and packet captures were analysed to assess the effectiveness of the measures implemented and to pinpoint any existing vulnerabilities. The results were then used to develop actionable recommendations to enhance network resilience, strengthen access controls, improve monitoring, and ensure compliance with healthcare security and privacy standards.

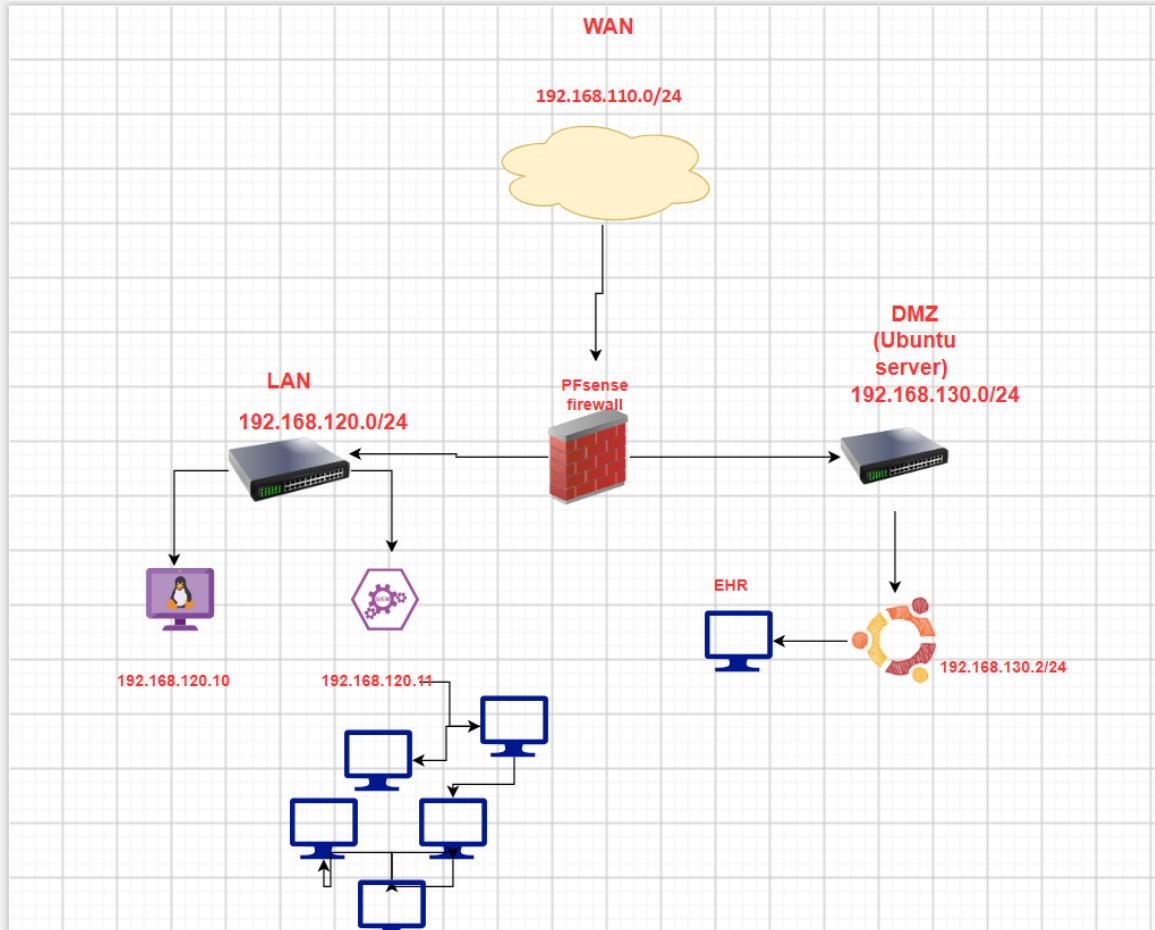
Phase-by-Phase Analysis

Phase 1: Network Design & Segmentation (Draw.io & pfSense)

Objective: Design a segmented network (WAN, LAN, DMZ) and implement access controls.

A virtual network was then deployed within Oracle VirtualBox, where pfSense was configured with separate WAN, LAN, and DMZ interfaces. Appropriate IP addressing and routing were established to ensure secure communication between authorised systems while maintaining isolation between network segments.

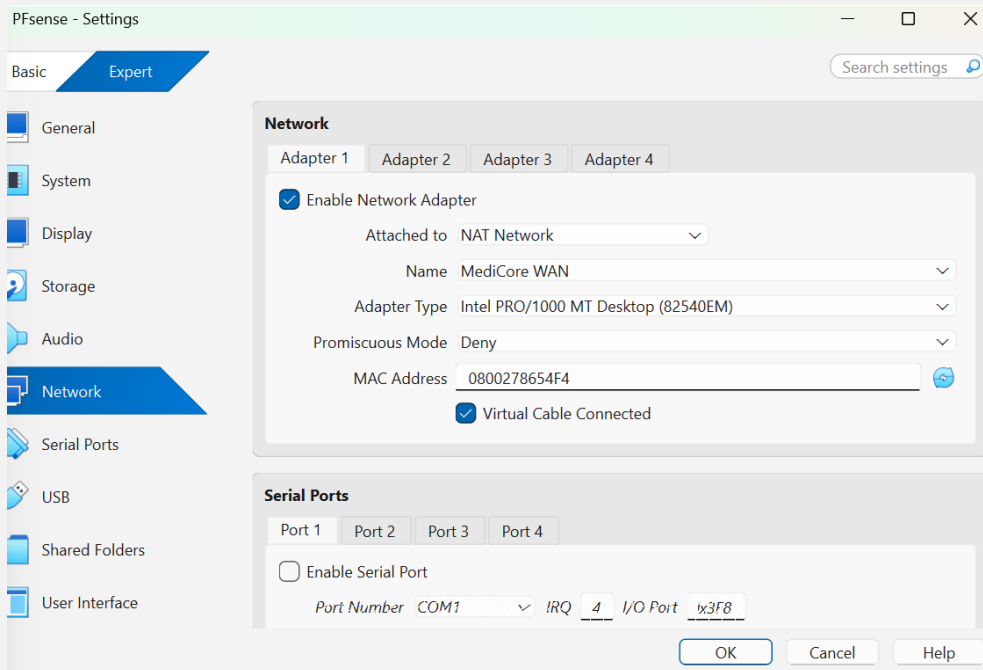
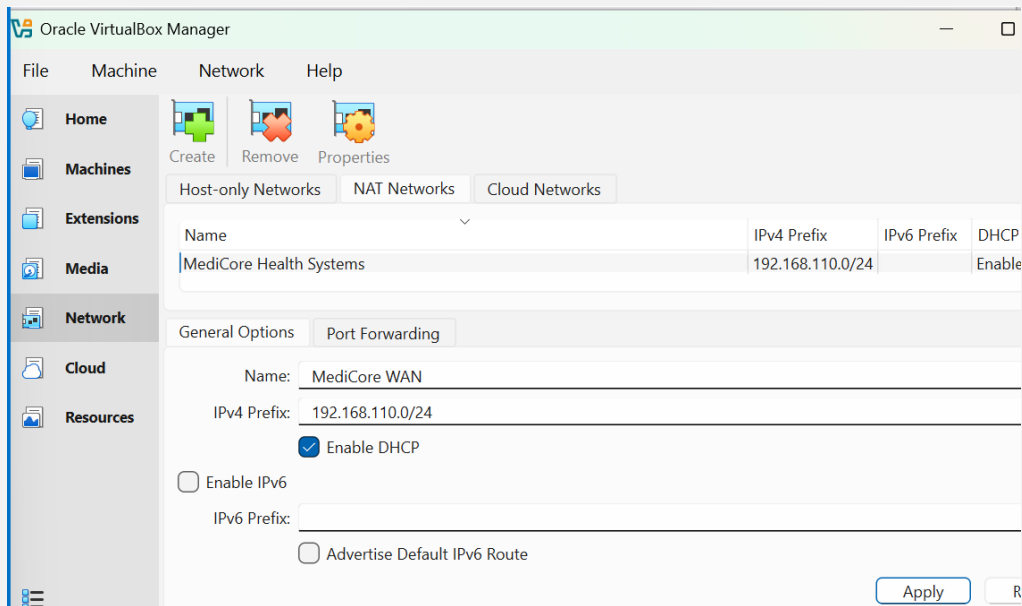
I implemented the following security controls: I segmented the WAN, LAN, and DMZ into separate network zones; isolated the patient portal from the internal clinical network; configured stateful packet inspection using pfSense firewall rules; enforced a default-deny firewall policy with explicitly permitted traffic; controlled routing between authorised network segments; restricted direct access to Electronic Health Record (EHR) systems; protected against unauthorised lateral movement between network zones; implemented Network Address Translation (NAT) for secure communication between internal and external networks; and limited administrative access to trusted management hosts.



Assigned IP addresses

WAN- Internet:	192.168.110.0/24
LAN	192.168.120.0/24
	Clinical Staff Workstations Administrative Workstations SOC Workstation Wazuh Server Medical Devices (IoMT)
DMZ	192.168.130.0/24 Patient Portal / Telehealth Server
Security Layer	pfSense Firewall

Attacker Network	Kali Linux
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192.168.110.3/wizard.php?xml=setup_wizard.xml&stepid=9

Kali DocsKali ForumsKali NetHunterExploit-DBGoogle Hacking DB

pfSense

SystemInterfacesFirewallServicesVPNStatusDiagnosticsHelp

WARNING:

The password for this account is insecure. Password is currently set to the default value (pfsense). Change the password as soon as possible.

Wizard / pfSense Setup / Wizard completed.

Step 9 of 9

Wizard completed.

Congratulations! pfSense is now configured.

We recommend that you check to see if there are any software updates available. Keeping your software up to date is one of the most important things you can do to maintain the security of your network.

Check for updates

Remember, we're here to help.

Click here to learn about Netgate 24/7/365 support services.

User survey

Please help all the people involved in improving and expanding pfSense software by taking a moment to answer this short survey (all answers are anonymous)

Anonymous User Survey

Useful resources.

- Learn more about Netgate's product line, services, and pfSense software from our website
- To learn about Netgate appliances and other offers, visit our store
- Become part of the pfSense community. Visit our forum
- Subscribe to our newsletter for ongoing product information, software announcements and special offers.

Finish

Status / Dashboard

+ ?

System Information

Name	Medicore.home.arpa
User	admin@192.168.110.4 (Local Database)
System	VirtualBox Virtual Machine Netgate Device ID: 5f42abf4164d1724da84
BIOS	Vendor: innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006 Boot Method: BIOS
Version	2.8.1-RELEASE (amd64) built on Wed Jul 1 15:25:00 UTC 2026 FreeBSD 15.0-CURRENT The system is on the latest version. Version information updated at Mon Jul 13 20:26:53 UTC 2026
CPU Type	Intel(R) Core(TM) 5 120U AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Disabled
MDS Mitigation	Inactive
Uptime	00 Hour 04 Minutes 16 Seconds
Current date/time	Mon Jul 13 20:30:12 UTC 2026
DNS server(s)	127.0.0.1 ::1

Netgate Services And Support

Contract Type

Community Support
Community Support Only

NETGATE & PFSense COMMUNITY SUPPORT RESOURCES

If you purchased your pfSense gateway firewall appliance from Netgate and elected **Community Support** at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the **NETGATE RESOURCE LIBRARY**.

You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.

- Upgrade Your Support
- Netgate Global Support FAQ
- Netgate Professional Services
- Community Support Resources
- Official pfSense Training by Netgate
- Visit Netgate.com

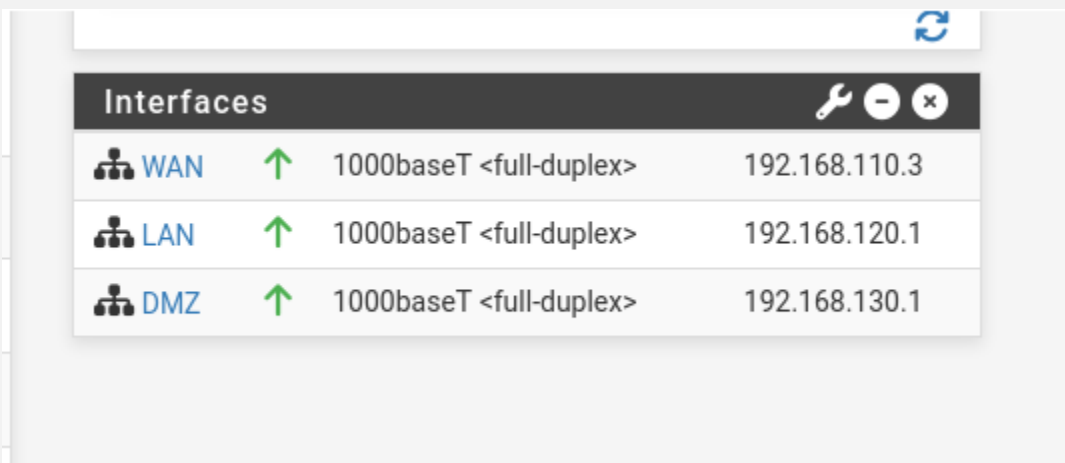
If you decide to purchase a Netgate Global TAC Support subscription, you **MUST** have your **Netgate Device ID (NDI)** from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase a TAC support plan [here](#).

Interfaces

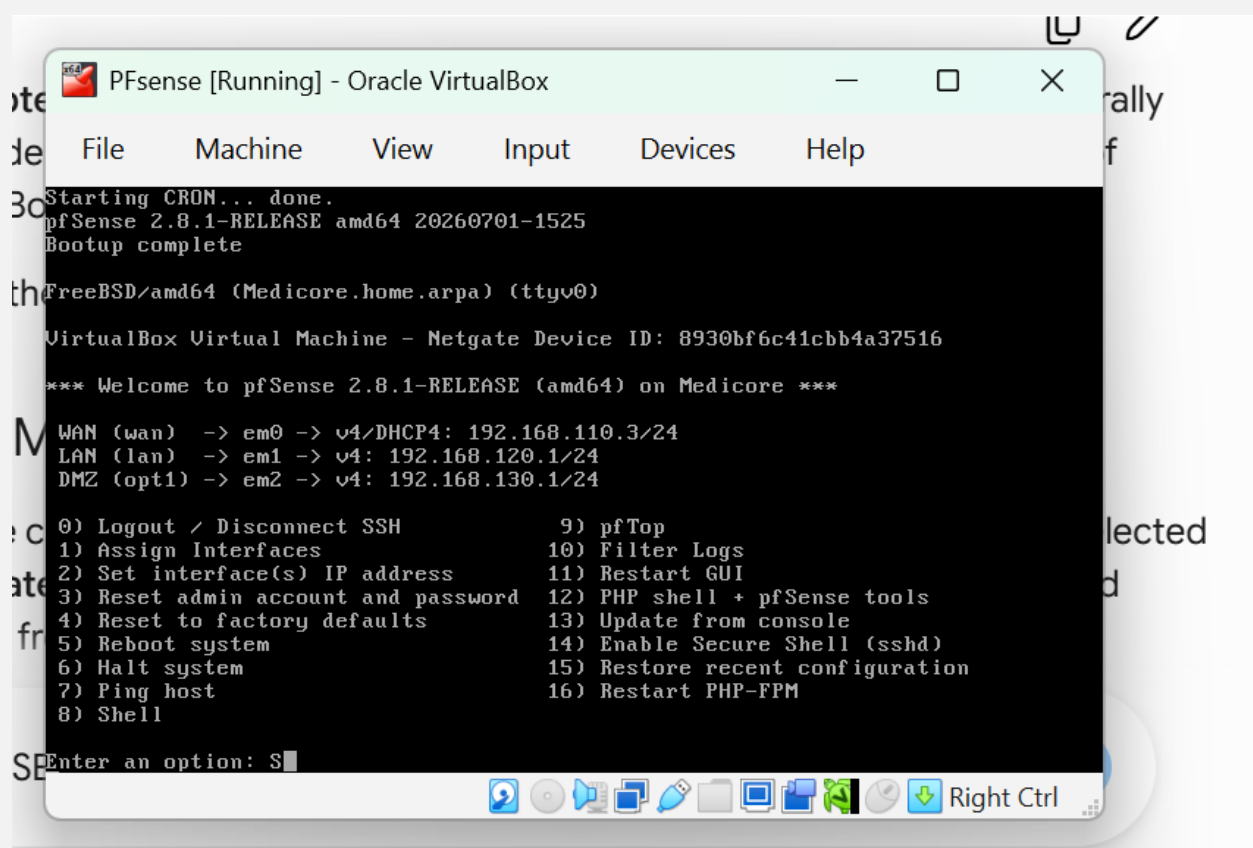
WAN

↑ 1000baseT <full-duplex>

192.168.110.3



Interfaces			
WAN	↑	1000baseT <full-duplex>	192.168.110.3
LAN	↑	1000baseT <full-duplex>	192.168.120.1
DMZ	↑	1000baseT <full-duplex>	192.168.130.1



```

PFsense [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260701-1525
Bootup complete
FreeBSD/amd64 (Medicore.home.arpa) (ttyv0)
VirtualBox Virtual Machine - Netgate Device ID: 8930bf6c41cbb4a37516
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on Medicore ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.110.3/24
LAN (lan) -> em1 -> v4: 192.168.120.1/24
DMZ (opt1) -> em2 -> v4: 192.168.130.1/24

0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell
9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM

Enter an option: S

```

Phase 2: Attack Simulation & Traffic Capture (Kali Linux & Wireshark)

Kali Linux is the simulated attacker platform to conduct controlled security tests against the segmented network environment. The activities were designed to represent common threats faced by healthcare organisations, including reconnaissance activities

and attempts to gain unauthorised access to protected systems. SSH was configured on Kali and Ubuntu, target user accounts were created, and I stimulated a realistic attack chain using Nmap and Kali from Kali Linux. Wireshark captured and analysed network traffic at specific monitoring locations. Packet captures were analysed to detect atypical communication behaviours, authentication attempts, and possibly harmful sessions. Relevant packets, connection information, and traffic patterns were gathered to aid in further analysis and offer proof of the simulated attacks.

The gathered information was analysed alongside firewall activity and security monitoring data to assess if the applied controls effectively limited unauthorised actions and produced suitable detection events.

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.120.10/24 brd 192.168.120.255 scope global dynamic noprefixroute eth0
        valid_lft 7171sec preferred_lft 7171sec
    inet6 fe80::5ae7:335f:aca1:1d05/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$
```

```
WAZUH Open Source Security Platform
https://wazuh.com
[wazuh-user@wazuh-server ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ab:31:77 brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 192.168.120.11/24 metric 1024 brd 192.168.120.255 scope global dynamic eth0
        valid_lft 7154sec preferred_lft 7154sec
    inet6 fe80::a00:27ff:feab:3177/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]# ss
```

```
ubuntu@ubuntu-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ce:96:ec brd ff:ff:ff:ff:ff:ff
    inet 192.168.130.2/24 brd 192.168.130.255 scope global dynamic noprefixroute enp0s3
        valid_lft 6468sec preferred_lft 6468sec
    inet6 fe80::a00:27ff:fece:96ec/64 scope link
        valid_lft forever preferred_lft forever

ubuntu@ubuntu-VirtualBox:~$ ping 192.168.120.10
PING 192.168.120.10 (192.168.120.10) 56(84) bytes of data.
```

Floating

WAN

LAN

DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	5/6 KiB	IPv4 *	DMZ subnets	*	*	*	none			    

↑ Add

↓ Add

 Delete

 Toggle

 Copy

 Save

 + Se



```

ubuntu@ubuntu-VirtualBox:~$ ping 192.168.120.10
PING 192.168.120.10 (192.168.120.10) 56(84) bytes of data.
64 bytes from 192.168.120.10: icmp_seq=1 ttl=63 time=5.49 ms
64 bytes from 192.168.120.10: icmp_seq=2 ttl=63 time=3.99 ms
64 bytes from 192.168.120.10: icmp_seq=3 ttl=63 time=3.53 ms
64 bytes from 192.168.120.10: icmp_seq=4 ttl=63 time=3.80 ms
64 bytes from 192.168.120.10: icmp_seq=5 ttl=63 time=4.06 ms
64 bytes from 192.168.120.10: icmp_seq=6 ttl=63 time=2.19 ms
^C
--- 192.168.120.10 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5014ms
rtt min/avg/max/mdev = 2.192/3.842/5.492/0.967 ms
ubuntu@ubuntu-VirtualBox:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=62 time=42.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=62 time=36.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=62 time=42.1 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=62 time=31.7 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=62 time=52.5 ms
^C
--- 8.8.8.8 ping statistics ---

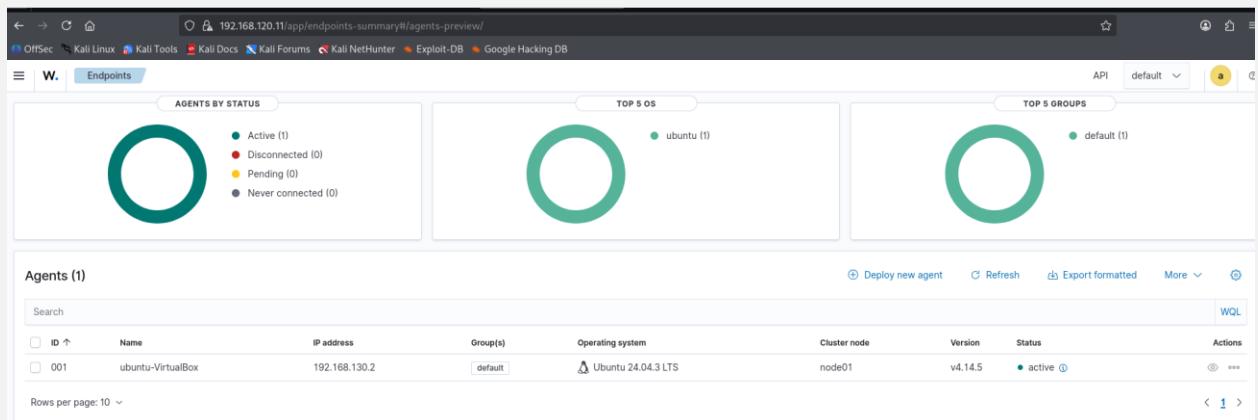
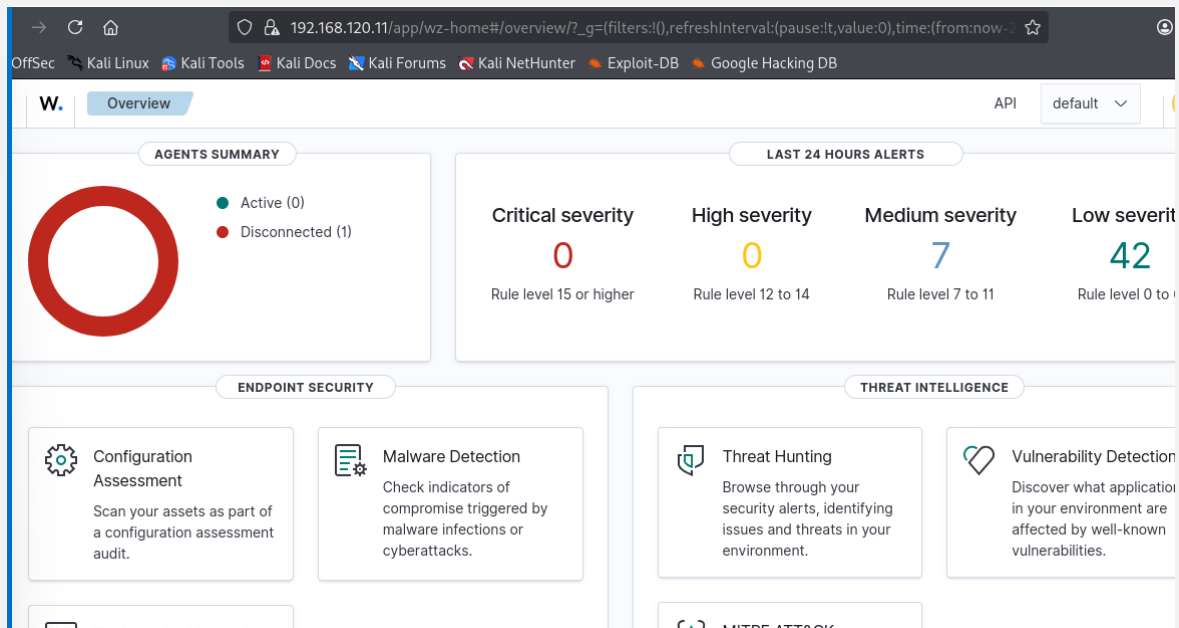
```

```

[root@wazuh-server ~]# sudo systemctl status wazuh-manager
■ wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-07-13 21:43:03 UTC; 1h 14min ago
     Process: 2039 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/
   Tasks: 231 (limit: 4646)
   Memory: 966.0M
     CPU: 46min 2.010s
   CGroup: /system.slice/wazuh-manager.service
           └─2303 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             └─2304 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
               └─2305 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                 └─2308 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                   └─2311 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                     └─2357 /var/ossec/bin/wazuh-authd
                       └─2378 /var/ossec/bin/wazuh-db
                         └─2433 /var/ossec/bin/wazuh-execd
                           └─2452 /var/ossec/bin/wazuh-analysisd
                             └─2473 /var/ossec/bin/wazuh-syscheckd
                               └─2493 /var/ossec/bin/wazuh-remoted
                                 └─2537 /var/ossec/bin/wazuh-logcollector
                                   └─2610 /var/ossec/bin/wazuh-monitord
                                     └─2627 /var/ossec/bin/wazuh-modulesd

Jul 13 21:42:58 wazuh-server env[2039]: wazuh-logcollector: Process 2568 not used by Wazuh, removing...
Jul 13 21:42:59 wazuh-server env[2039]: Started wazuh-logcollector...
Jul 13 21:42:59 wazuh-server env[2039]: wazuh-monitord: Process 2582 not used by Wazuh, removing...
Jul 13 21:43:00 wazuh-server env[2039]: Started wazuh-monitord...

```



Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

S

No.	Time	Source	Destination	Protocol	Length	Info
17	10.084298891	192.168.120.10	192.168.110.3	TLSv1.2	143	App
18	10.084427872	192.168.120.10	192.168.110.3	TLSv1.2	192	App
19	10.090190552	192.168.110.3	192.168.120.10	TCP	66	443
20	10.090191205	192.168.110.3	192.168.120.10	TCP	66	443
21	10.090191340	192.168.110.3	192.168.120.10	TLSv1.2	101	App
22	10.090241355	192.168.120.10	192.168.110.3	TLSv1.2	512	App
23	10.090413252	192.168.120.10	192.168.110.3	TCP	66	400
24	10.093946588	192.168.110.3	192.168.120.10	TCP	66	443
25	10.099209423	192.168.110.3	192.168.120.10	TLSv1.2	114	App
26	10.099260745	192.168.120.10	192.168.110.3	TCP	66	400

Frame 1: Packet, 143 bytes on wire (114 bytes captured) on interface eth0

- Ethernet II, Src: PCSSystemtec_8a:35:d2, Dst: 08:00:27:83:33:b1
- Internet Protocol Version 4, Src: 192.168.120.10, Dst: 192.168.110.3
- Transmission Control Protocol, Src Port: 443, Dst Port: 4000
- Transport Layer Security

"S" is not a valid protocol or protocol field. Packets: 26 Profile: Default

```

ubuntu@ubuntu-VirtualBox:~$ sudo systemctl start ssh
sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
ubuntu@ubuntu-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-14 22:11:49 WAT; 18s ago
 TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 16692 (sshd)
      Tasks: 1 (limit: 3680)
     Memory: 1.2M (peak: 1.4M)
        CPU: 37ms
     CGroup: /system.slice/ssh.service
            └─16692 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 14 22:11:49 ubuntu-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jul 14 22:11:49 ubuntu-VirtualBox sshd[16692]: Server listening on 0.0.0.0 port 22.
Jul 14 22:11:49 ubuntu-VirtualBox sshd[16692]: Server listening on :: port 22.
Jul 14 22:11:49 ubuntu-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

```

```

(kali㉿kali)-[~]
└─$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-14 16:42:03 EDT; 34min ago
 Invocation: 26cef7c5e1a4480ba434eb174a42910a
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 618 (sshd)
      Tasks: 1 (limit: 2118)
     Memory: 1.3M (peak: 2.9M, swap: 788K, swap peak: 788K)
        CPU: 61ms
     CGroup: /system.slice/ssh.service
            └─618 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 14 16:42:03 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jul 14 16:42:03 kali sshd[618]: Server listening on 0.0.0.0 port 22.
Jul 14 16:42:03 kali sshd[618]: Server listening on :: port 22.
Jul 14 16:42:03 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

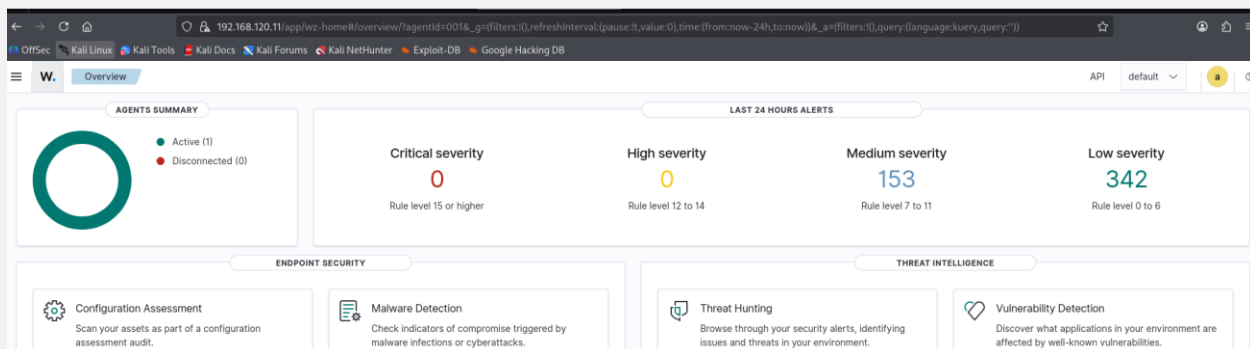
```

Phase 3: Security Event Monitoring & Response (Wazuh SIEM)

Wazuh Manager was deployed on the internal LAN environment using an Ubuntu Server platform to provide centralised monitoring and event analysis. Wazuh agent was installed on selected systems, including DMZ servers and endpoints, to gather

security-related data like authentication events, system modifications, and signs of suspicious activity. After deployment, monitoring rules were configured to detect possible threats associated with the healthcare setting. These encompassed ongoing authentication errors, network scanning actions, unauthorised access attempts, and privilege escalation activities. The alert configuration was modified to prioritise important events while minimising excessive notifications.

I evaluated active response capabilities to determine how automated actions could support incident containment. I considered appropriate response mechanisms, such as blocking suspicious IP addresses and executing predefined security scripts, to reduce the impact of detected threats and improving response efficiency. I reviewed the generated alerts alongside evidence collected from firewall logs and Wireshark packet captures, allowing me to correlate security events from multiple perspectives and demonstrate the workflow.



```
root@ubuntu-VirtualBox:/home/ubuntu# adduser adminmedi
info: Adding user `adminmedi' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `adminmedi' (1003) ...
info: Adding new user `adminmedi' (1003) with group `adminmedi (1003)' ...
info: Creating home directory `/home/adminmedi' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for adminmedi
Enter the new value, or press ENTER for the default
  Full Name []: JFORJUMY
```

```
root@ubuntu-VirtualBox:/home/ubuntu# adduser medicore
info: Adding user `medicore' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `medicore' (1004) ...
info: Adding new user `medicore' (1004) with group `medicore (1004)' ...
info: Creating home directory `/home/medicore' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for medicore
Enter the new value, or press ENTER for the default
  Full Name []:
```

```
  Other []:
Is the information correct? [Y/n] y
info: Adding new user `medicore' to supplemental / extra groups `users' ...
info: Adding user `medicore' to group `users' ...
root@ubuntu-VirtualBox:/home/ubuntu# echo "MediCore Health Systems is a rapidly growing healthcare provider"> /home/adminmedi/secret.txt
root@ubuntu-VirtualBox:/home/ubuntu# cd /home/adminmedi
root@ubuntu-VirtualBox:/home/adminmedi# ls
secret.txt
root@ubuntu-VirtualBox:/home/adminmedi# cat secret.txt
MediCore Health Systems is a rapidly growing healthcare provider
root@ubuntu-VirtualBox:/home/adminmedi#
```

```
ratio_etc forever preferred_etc forever
main PID: 618 (sshd)

(kali@kali)-[~]
$ nmap 192.168.120.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:05 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.016s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
systemd[1]: Server listening on 0.0.0.0 port 22.
systemd[1]: Server listening on 0.0.0.0 port 22.
systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.120.11
Host is up (0.014s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
443/tcp   open  https
MAC Address: 08:00:27:AB:31:77 (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.120.10
Host is up (0.0025s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
system, software, processes, and network
to detect misconfigurations, unauthorized

Nmap done: 256 IP addresses (3 hosts up) scanned in 13.72 seconds
```

I gathered information using the nmap reconnaissance on the network to attack by using the Ip address 192.168.120.0/24 and it was used to correlate the Ip on the design segmentation on kali machine under LAN and firewall. I scanned the networks to know which hosts are alive

```

(kali㉿kali)-[~]
$ nmap -sn 192.168.120.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:14 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0059s latency).
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.120.11
Host is up (0.0072s latency).
MAC Address: 08:00:27:AB:31:77 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.120.10
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.07 seconds
(kali㉿kali)-[~]
$ nmap -sV 192.168.120.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:16 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0043s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  ssl/http
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.34 seconds
(kali㉿kali)-[~]
$ sS

```

```

(kali㉿kali)-[~]
$ nmap -sS 192.168.120.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:20 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0070s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 5.53 seconds

```

```

(kali@kali)-[~]
$ nmap -sV 192.168.120.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:24 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0054s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  Unbound
80/tcp    open  http    nginx
443/tcp   open  ssl/http nginx
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.49 seconds

(kali@kali)-[~]
$ nmap -sU 192.168.120.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:31 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0066s latency).
Not shown: 997 open/filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
123/udp   open  ntp
161/udp   open  snmp
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.66 seconds

```

SNMP shows interface names, Ip addresses and descriptions.

The left screenshot shows the pfSense GUI with the following interface assignments:

- WAN (wan) -> em0 -> v4/DHCP4: 192.168.110.3/24
- LAN (lan) -> em1 -> v4: 192.168.120.1/24
- DMZ (opt1) -> em2 -> v4: 192.168.130.1/24

The right screenshot shows a terminal window with the following output from an nmap -sU scan of 192.168.120.1:

```

(kali@kali)-[~]
$ nmap -sU 192.168.120.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:34 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0051s latency).
PORT      STATE SERVICE
161/udp   open  snmp

```

```

status ssh
(kali㉿kali)-[~]
$ nmap 192.168.130.0/24
lib/systemd/system/ssh.service: enabled; preset: disabled
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:39 -0400 14min ago
Nmap scan report for 192.168.130.1 (174a42910a)
Host is up (0.011s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http  peak: 2.9M, swap: 788K, swap peak: 788K
443/tcp   open  https
CGroup: /system.slice/ssh.service
Nmap scan report for 192.168.130.2
Host is up (0.011s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
22/tcp    open  ssh  ali sshd[618]: Server listening on 0.0.0.0 port 22.
22/tcp    open  ssh  ali sshd[618]: Server listening on :: port 22.
Jul 14 16:42:03 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Nmap done: 256 IP addresses (2 hosts up) scanned in 12.60 seconds

```

```

(kali㉿kali)-[~]
$ nano password.txt

(kali㉿kali)-[~]
$ cat users.txt
adminmedi
ubuntu
Medicore
JUmy

```

```

(kali㉿kali)-[~]
$ cat users.txt
adminmedi
ubuntu
Medicore
JUmy

(kali㉿kali)-[~]
$ cat password.txt
password
ubuntu
123456789
password

(kali㉿kali)-[~]
$

```

```

(kali@kali)-[~]
$ hydra -L users.txt -P password.txt -t4 -V ssh://192.168.130.2
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-14 19:28:52
[DATA] max 4 tasks per 1 server, overall 4 tasks, 16 login tries (l:4/p:4), ~4 tries per task
[DATA] attacking ssh://192.168.130.2:22/
[ATTEMPT] target 192.168.130.2 - login "adminmedi" - pass "password" - 1 of 16 [child 0] (0/0)
[ATTEMPT] target 192.168.130.2 - login "adminmedi" - pass "ubuntu" - 2 of 16 [child 1] (0/0)
[ATTEMPT] target 192.168.130.2 - login "adminmedi" - pass "123456789" - 3 of 16 [child 2] (0/0)
[ATTEMPT] target 192.168.130.2 - login "adminmedi" - pass "password" - 4 of 16 [child 3] (0/0)
[22][ssh] host: 192.168.130.2 login: adminmedi password: password
[ATTEMPT] target 192.168.130.2 - login "ubuntu" - pass "password" - 5 of 16 [child 3] (0/0)
[22][ssh] host: 192.168.130.2 login: adminmedi password: password
[ATTEMPT] target 192.168.130.2 - login "ubuntu" - pass "ubuntu" - 6 of 16 [child 0] (0/0)
[22][ssh] host: 192.168.130.2 login: ubuntu password: ubuntu
[ATTEMPT] target 192.168.130.2 - login "Medicore" - pass "password" - 9 of 16 [child 0] (0/0)
[ATTEMPT] target 192.168.130.2 - login "Medicore" - pass "ubuntu" - 10 of 16 [child 1] (0/0)
[ATTEMPT] target 192.168.130.2 - login "Medicore" - pass "123456789" - 11 of 16 [child 2] (0/0)
[ATTEMPT] target 192.168.130.2 - login "Medicore" - pass "password" - 12 of 16 [child 3] (0/0)
[ATTEMPT] target 192.168.130.2 - login "JUmy" - pass "password" - 13 of 16 [child 0] (0/0)
[ATTEMPT] target 192.168.130.2 - login "JUmy" - pass "ubuntu" - 14 of 16 [child 1] (0/0)
[ATTEMPT] target 192.168.130.2 - login "JUmy" - pass "123456789" - 15 of 16 [child 2] (0/0)
[ATTEMPT] target 192.168.130.2 - login "JUmy" - pass "password" - 16 of 16 [child 3] (0/0)
1 of 1 target successfully completed, 3 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-14 19:29:07
(kali@kali)-[~]

```

```

(kali@kali)-[~]
$ ssh adminmedi@192.168.130.2
The authenticity of host '192.168.130.2 (192.168.130.2)' can't be established.
ED25519 key fingerprint is: SHA256:tt5qRkQqf6Sm0o/vgrJnwJbYuX+Y9mzyj1WZS5gj/Vk
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.130.2' (ED25519) to the list of known hosts.
adminmedi@192.168.130.2's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-34-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

445 updates can be applied immediately.
312 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

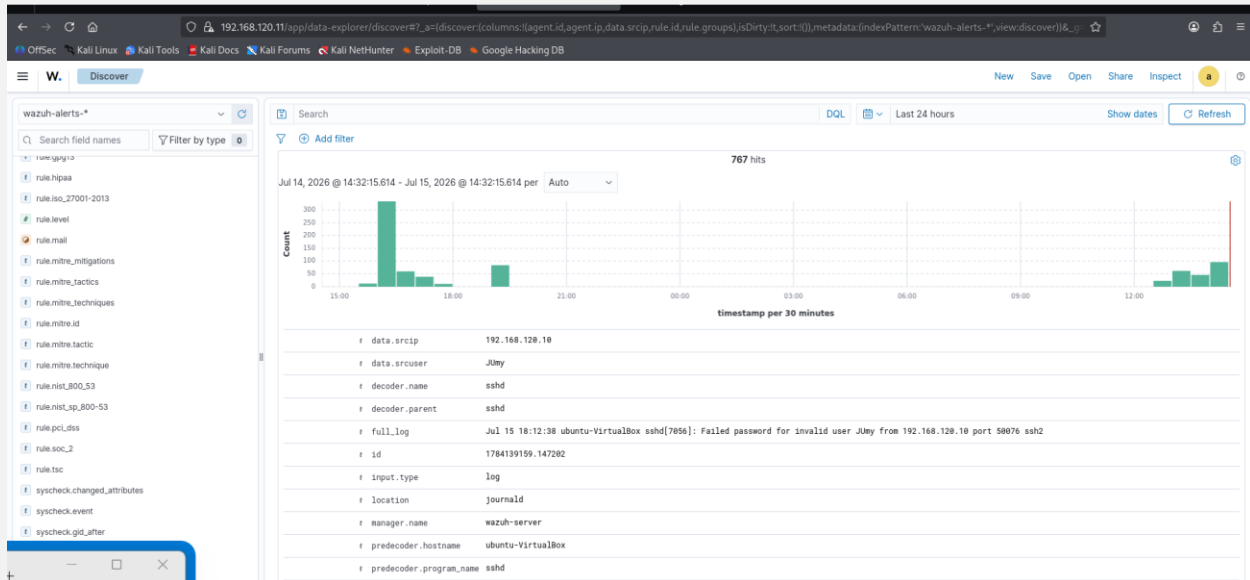
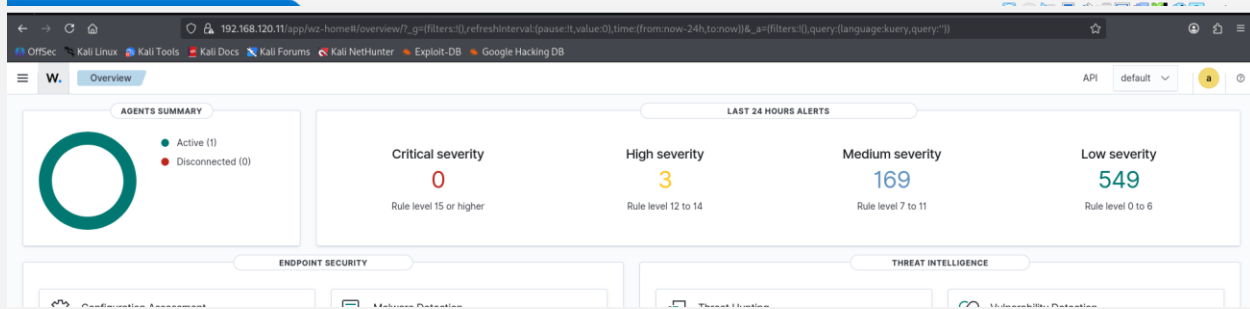
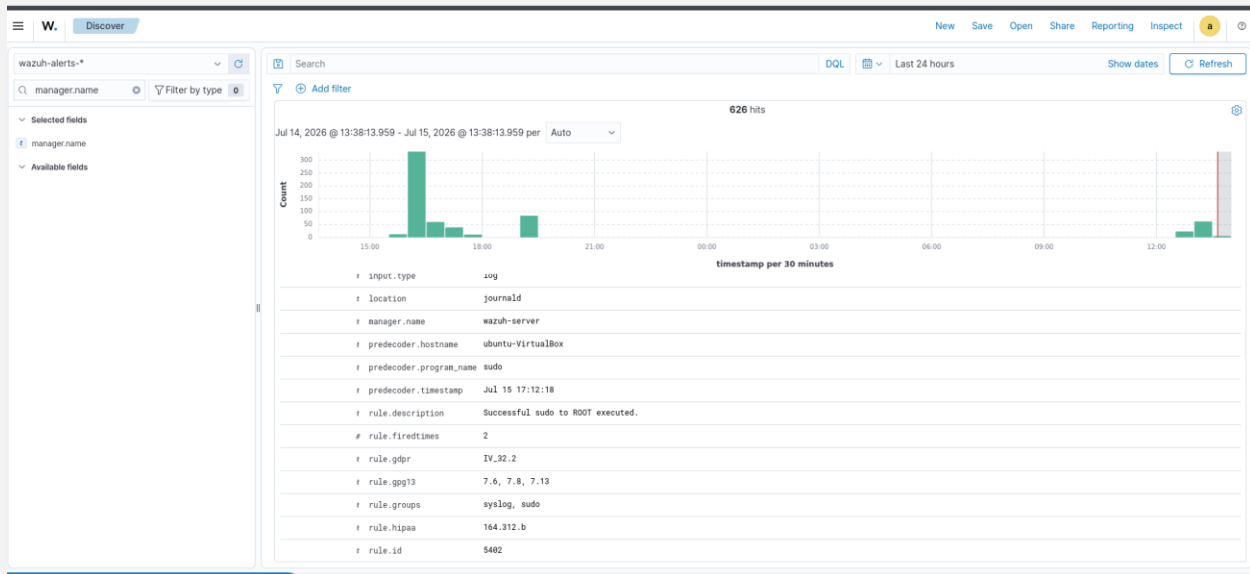
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

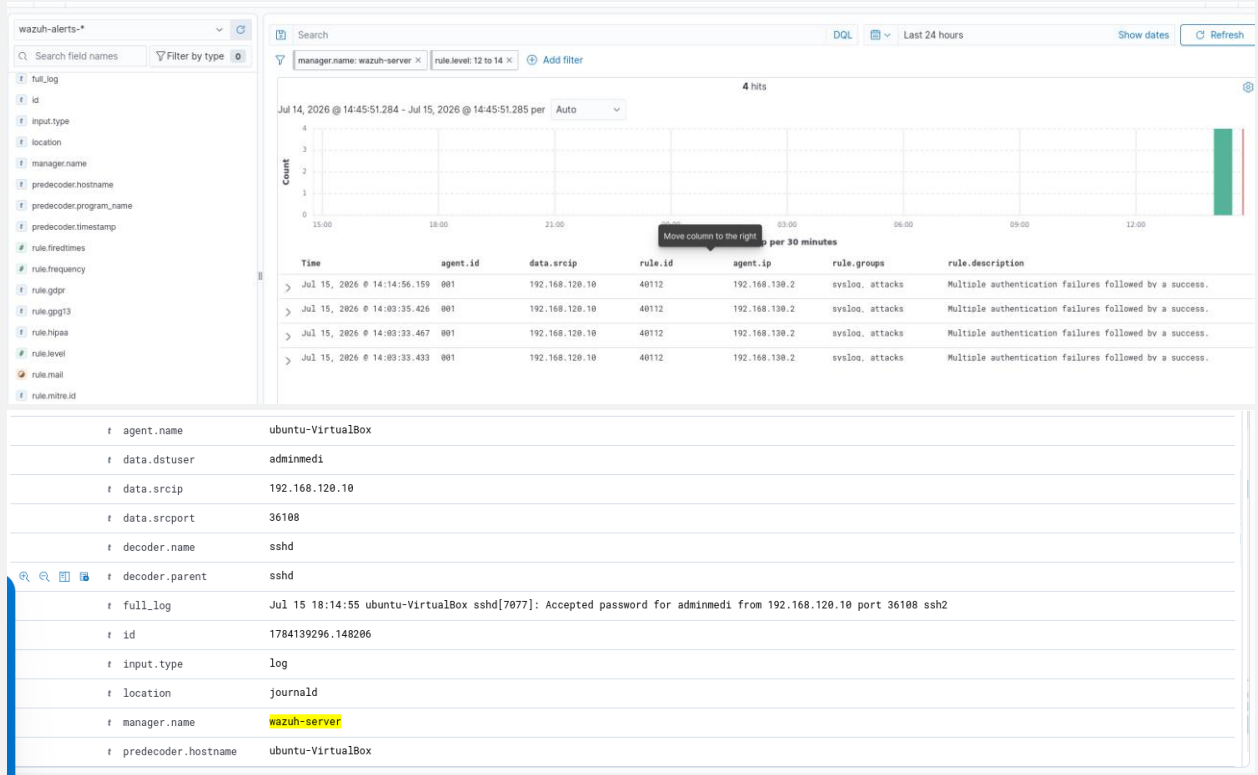
```

```

adminmedi@ubuntu-VirtualBox:~$ ls
secret.txt  snap
adminmedi@ubuntu-VirtualBox:~$ cat secret.txt
MediCore Health Systems is a rapidly growing healthcare provider

```





No.	Time	Source	Destination	Protocol	Length	Info
381	25.287468783	192.168.120.10	192.168.130.2	TCP	74	50100 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2147751180 TSecr=0 WS=256
382	25.292633565	192.168.130.2	192.168.120.10	TCP	66	22 → 50060 [FIN, ACK] Seq=1804 Ack=1313 Win=64256 Len=0 TSval=1241753229 TSecr=2147751128
383	25.292698825	192.168.120.10	192.168.130.2	TCP	66	50060 → 22 [ACK] Seq=1313 Ack=1805 Win=66048 Len=0 TSval=2147751185 TSecr=1241753229
384	25.299578295	192.168.130.2	192.168.120.10	TCP	66	22 → 50046 [ACK] Seq=1804 Ack=1312 Win=64256 Len=0 TSval=1241753235 TSecr=2147751150
385	25.319880750	192.168.130.2	192.168.120.10	TCP	66	22 → 50086 [ACK] Seq=1 Ack=24 Win=65152 Len=0 TSval=1241753250 TSecr=2147751169
386	25.319888842	192.168.130.2	192.168.120.10	TCP	74	22 → 50100 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=1241753258 TSecr=2147751180 WS=128
387	25.319927124	192.168.120.10	192.168.130.2	TCP	66	50100 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2147751212 TSecr=1241753258
388	25.327957322	192.168.120.10	192.168.130.2	SSHv2	89	Client: Protocol (SSH-2.0-libssh_0.11.3)
389	25.340719844	192.168.130.2	192.168.120.10	SSHv2	118	Server: Encrypted packet (len=52)
390	25.344183967	192.168.130.2	192.168.120.10	TCP	66	22 → 50046 [ACK] Seq=1804 Ack=1313 Win=64256 Len=0 TSval=1241753282 TSecr=2147751157
391	25.346388857	192.168.120.10	192.168.130.2	SSHv2	118	Client: Encrypted packet (len=52)
392	25.346774254	192.168.120.10	192.168.130.2	TCP	66	50074 → 22 [FIN, ACK] Seq=1312 Ack=1804 Win=66048 Len=0 TSval=2147751239 TSecr=1241753278
393	25.347595325	192.168.120.10	192.168.130.2	TCP	74	50100 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2147751240 TSecr=0 WS=256
394	25.409957826	192.168.120.10	192.168.130.2	TCP	66	[TCP Retransmission] 50074 → 22 [FIN, ACK] Seq=1312 Ack=1804 Win=66048 Len=0 TSval=2147751302 TSecr=1241753278
395	25.431373015	192.168.130.2	192.168.120.10	TCP	66	22 → 50100 [ACK] Seq=1 Ack=24 Win=65152 Len=0 TSval=1241753290 TSecr=2147751220
396	25.434783769	192.168.130.2	192.168.120.10	SSHv2	558	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=284)
397	25.434785101	192.168.130.2	192.168.120.10	TCP	66	22 → 50046 [FIN, ACK] Seq=1804 Ack=1313 Win=64256 Len=0 TSval=1241753306 TSecr=2147751157
398	25.436577922	192.168.120.10	192.168.130.2	TCP	66	50046 → 22 [ACK] Seq=1313 Ack=1805 Win=66048 Len=0 TSval=2147751329 TSecr=1241753306
399	25.440215100	192.168.120.10	192.168.130.2	SSHv2	82	Client: New Keys
400	25.444565518	192.168.130.2	192.168.120.10	TCP	66	22 → 50074 [ACK] Seq=1804 Ack=1312 Win=64256 Len=0 TSval=1241753378 TSecr=2147751239
401	25.448559629	192.168.130.2	192.168.120.10	TCP	74	22 → 50100 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=1241753391 TSecr=2147751240 WS=128
402	25.448598155	192.168.120.10	192.168.130.2	TCP	66	50100 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2147751341 TSecr=1241753381
403	25.449348002	192.168.120.10	192.168.130.2	SSHv2	89	Client: Protocol (SSH-2.0-libssh_0.11.3)

192.168.120.11/app/dashboards-settings#/manager/?tab=configu
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

W. Settings

Manager configuration Refresh Save Restart Manager

Edit ossec.conf of Manager

```

235 | <timeout_allowed>yes</timeout_allowed>
236 | </command>
237 |
238 | <ossec_config>
239 | <active-response>
240 |   <disabled>no</disabled>
241 |   <command>firewall-drop</command>
242 |   <location>local</location>
243 |   <rules_id>5763</rules_id>
244 |   <timeout>180</timeout>
245 | </active-response>
246 | </ossec_config>

```

Manager configuration Refresh Save Restart Manager

Edit ossec.conf of Manager

```

235 | <timeout_allowed>yes</timeout_allowed>
236 | </command>
237 |
238 |
239 | <active-response>
240 |   <disabled>no</disabled>
241 |   <command>firewall-drop</command>
242 |   <location>local</location>
243 |   <rules_id>5763</rules_id>
244 |   <timeout>180</timeout>
245 | </active-response>
246 |

```

< Manager configuration

↻ Refresh

Edit **ossec.conf** of **Manager**

```
235 | <timeout_allowed>yes</timeout_allowed>  
236 | </command>  
237 |  
238 |  
239 | <active-response>  
240 |   <disabled>no</disabled>  
241 |   <command>firewall-drop</command>  
242 |   <location>local</location>  
243 |   <rules_group>sshd</rules_group>  
244 |   <timeout>0</timeout>  
245 | </active-response>  
246 |  
247 |  
248 | <!-- Log analysis -->  
249 | <localfile>  
250 |   <log_format>command</log_format>  
251 |   <command>df -P</command>
```

Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-15 15:45:46

(kali@kali)-[~]

\$ hydra -L users.txt -P password.txt -t4 -V ssh://192.168.130.11

Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-15 15:45:58

[DATA] max 4 tasks per 1 server, overall 4 tasks, 16 login tries (l:4/p:4), ~4 tries per task

[DATA] attacking ssh://192.168.130.11:22/

[ERROR] could not connect to ssh://192.168.130.11:22 - Timeout connecting to 192.168.130.11



MITRE ATT&CK

Firewall / Rules / LAN

FloatingWANLANDMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	✓ 0/0 B	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☐	✗ 0/3 KiB	IPv4 *	LAN subnets	*	DMZ subnets	*	*	none			
☐	✓ 6/781.98 MiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
☐	✓ 0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

Add

Add

Delete

Toggle

Copy

Save

Separator

Final Findings, Impact and Recommendation.

The security assessment of Medicare Health Systems identified several areas where improvements were required to strengthen the organisation's cybersecurity posture. I found out that the original network design lacked sufficient segmentation between critical environments, allowing unnecessary communication pathways user workstations, clinical systems, administrative resources, and medical devices. Another thing found is that limited monitoring capabilities further reduced the organisation's ability to quickly identify suspicious behaviour and respond to potential security incidents. After the deployment of the segmented network design, firewall measures, and centralised monitoring, the environment showed enhanced visibility and decreased risk. Simulated attacks with Kali Linux, aided by network analysis through Wireshark, effectively produced security incidents that Wazuh identified. The findings verified that enhanced segmentation and monitoring controls decreased unauthorised access attempts and restricted the capacity of threats between network zones.

The highest priority should be to further enhance technical controls that protect critical healthcare systems and patient data. Firewall policies should continue to follow the principle of least privilege, ensuring that communication is only permitted where there is a clear operational requirement. Administrative services like Secure Shell (SSH) need to be strengthened by turning off direct root logins, mandating key-based authentication, limiting access to trusted management systems, and modifying default settings as necessary. Moreover, disabling unneeded network ports and services is essential to minimise the attack surface, while implementing multi-factor

authentication for privileged accounts and remote administrative access is crucial. Periodic assessments of firewall rules and access control lists must be performed to eliminate outdated or excessively permissive settings. Security logs must be maintained for a suitable duration to assist with forensic investigation, regulatory adherence, and auditing needs. Regular vulnerability assessments, configuration reviews, and security audits must be integrated into the organisation's governance framework to detect weaknesses before they can be exploited.

Based on these findings, Medcore Health Systems should continue enhancing its cybersecurity controls by maintaining strict network segmentation to reduce the impact of potential security incidents, review and enforce access controls regularly by ensuring that only necessary communication and access are permitted, expand security monitoring capabilities by using centralised monitoring solutions to provide real-time visibility, alerting, and faster incident response. In addition to these, performing vulnerability testing, configuration reviews and simulating attack exercises will identify weaknesses before they can be exploited, improving awareness by provided ongoing training to employees will help focus on recognising threats such as unauthorised access attempts, unsafe security practices and phishing. Implementing these recommendations will help Medcore Health Systems reduce cyber risks, improve regulatory compliance, protect sensitive patient information, and support the continued delivery of safe and reliable healthcare services.

Conclusion

This project illustrates how an effectively structured Security Operations Centre (SOC) can greatly enhance an organisation's capacity to detect, examine, and address cybersecurity threats in a healthcare setting. Through the integration of network segmentation, firewall implementation, ongoing monitoring, and traffic analysis, the simulated infrastructure offered a realistic depiction of how layered security measures collaborate to mitigate organisational risk. The study revealed that dubious behaviour could be detected early, allowing for prompt evaluation and diminishing the chances of intruders accessing vital resources like the electronic health record (EHR) system.

The practical exercises reflected many of the responsibilities carried out by SOC analysts in real-world settings. Creating a secure network framework, establishing protective measures, observing security occurrences, scrutinising network data, and examining simulated breaches reflected the stages of an actual security incident. Employing industry-standard technologies such as pfSense, Wazuh, Wireshark, Kali Linux, Oracle VirtualBox, Ubuntu Server, and Draw.io offered significant practical experience while showcasing how various security solutions function collectively to enhance threat detection, incident response, and ongoing security enhancement. Even though the setting was artificial, the methods, procedures, and decision-making mirrored those applied in contemporary healthcare organisations.

The outcomes of this project highlight several important lessons for MediCore Health Systems. Strong network segmentation reduces the potential for lateral movement, carefully managed firewall policies limit unnecessary access to sensitive resources, and continuous monitoring improves visibility across the infrastructure. Equally important is the need for regular security assessments, ongoing review of security configurations, and staff awareness to ensure that technical controls remain effective as the organisation evolves. Utilising a proactive strategy for cybersecurity enhances the safeguarding of confidential patient data while aiding in regulatory adherence, operational robustness, and the seamless provision of healthcare services. This project shows that integrating preventative, detective, and responsive security measures creates a stronger basis for safeguarding essential healthcare systems from emerging cyber threats.

References

[SOC Analysis II Walkthrough.pdf - Google Drive](#)

[SOC_Analysis_III_Walkthrough.pdf - Google Drive](#)

[SOC Analysis 1 _ walkthrough.pdf - Google Drive](#)

<https://documentation.wazuh.com/current/user-manual/agent/agent-management/remove-agents/remove.html>

<https://documentation.wazuh.com/current/user-manual/capabilities/active-response/ar-use-cases/blocking-ssh-brute-force.html>

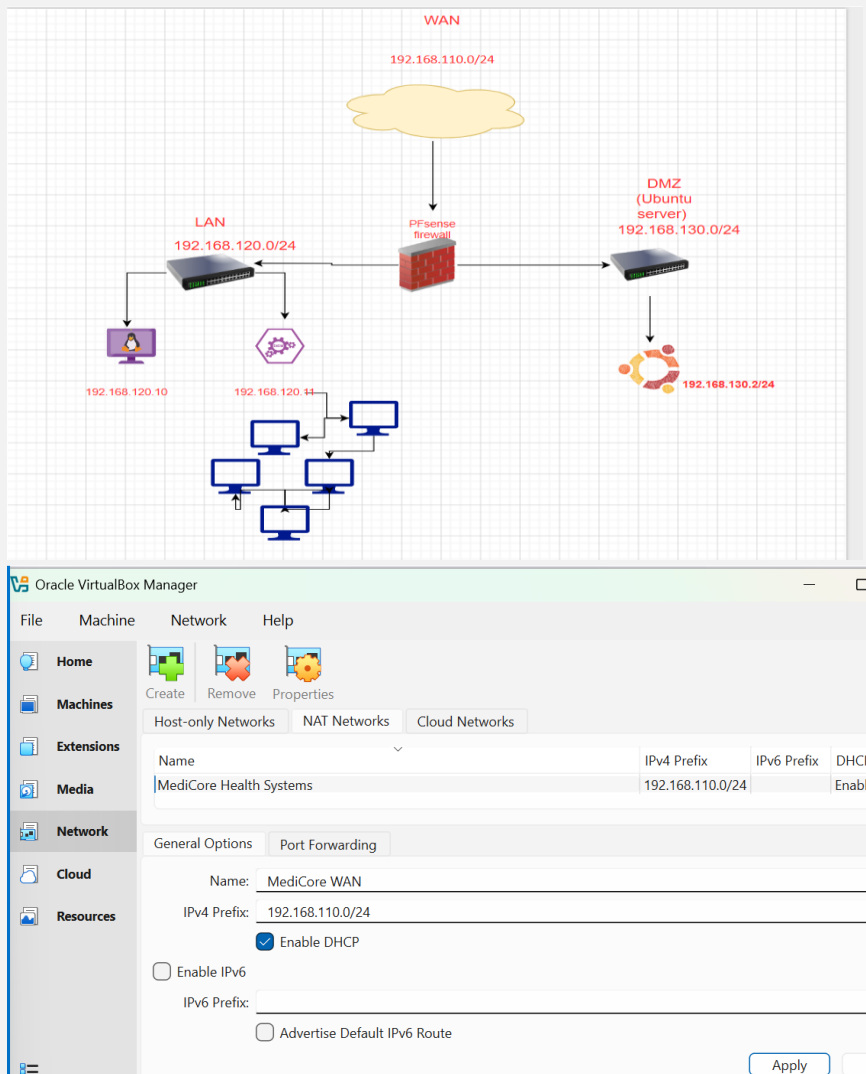
<https://documentation.wazuh.com/current/user-manual/capabilities/active-response/ar-use-cases/blocking-ssh-brute-force.html>

[My courses | 10Alytics](#)

<https://docs.netgate.com/pfsense/en/latest/index.html>

APPENDICES

Appendix A



```
PFsense [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Generating RRD graphs...done.
Starting syslog...done.
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260701-1525
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 5f42abf4164d1724da84

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.110.3/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults         13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: 
```

Floating WAN LAN **DMZ**

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	5/6 KiB	IPv4 *	DMZ subnets	*	*	*	*	none		

Add Add Delete Toggle Copy Save Add

Accessing Firewall Rules to configure traffic control between network zones. LAN allows rules to confirm that LAN hosts can access any destination while DMZ interface creates outbound traffic permissions.

Appendix B

```
ratio_err forever preferred_err forever
Main PID: 618 (sshd)

(kali@kali)-[~]
$ nmap 192.168.120.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:05 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.016s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.120.11
Host is up (0.014s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
443/tcp   open  https
MAC Address: 08:00:27:AB:31:77 (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.120.10
Host is up (0.0025s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
system, software, processes, and network
Nmap done: 256 IP addresses (3 hosts up) scanned in 13.72 seconds
```

SECURITY OPERATIONS

PCI DSS

Global security standard for entities that store, process, or transmit payment cardholder information

```

(kali㉿kali)-[~]
$ nmap -sn 192.168.120.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:14 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0059s latency).
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.120.11
Host is up (0.0072s latency).
MAC Address: 08:00:27:AB:31:77 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.120.10
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.07 seconds
(kali㉿kali)-[~]
$ nmap -sV 192.168.120.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:16 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0043s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  ssl/http
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.34 seconds
(kali㉿kali)-[~]
$ sS

```

```

(kali㉿kali)-[~]
$ nmap -sS 192.168.120.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:20 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0070s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 5.53 seconds

```

```

nmap done: 1 IP address (1 host up) scanned in 51.93 seconds

(kali㉿kali)-[~]
$ nmap -sV 192.168.120.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:24 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0054s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
53/tcp    open  domain Unbound
80/tcp    open  http   nginx
443/tcp   open  ssl/http nginx
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.49 seconds

```

```

(kali㉿kali)-[~]
$ nmap -sU 192.168.120.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 18:31 -0400
Nmap scan report for Medicore.home.arpa (192.168.120.1)
Host is up (0.0066s latency).
Not shown: 997 open/filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
123/udp   open  ntp
161/udp   open  snmp
MAC Address: 08:00:27:83:33:B1 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.66 seconds

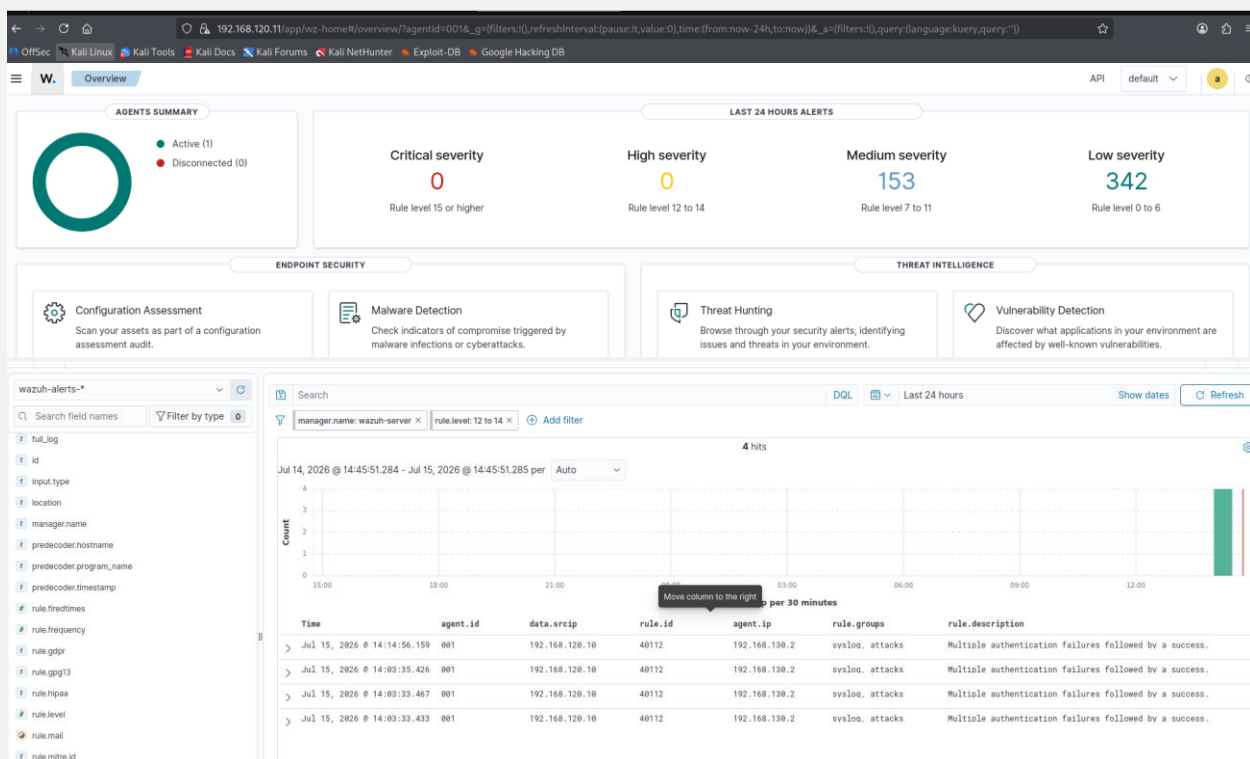
```

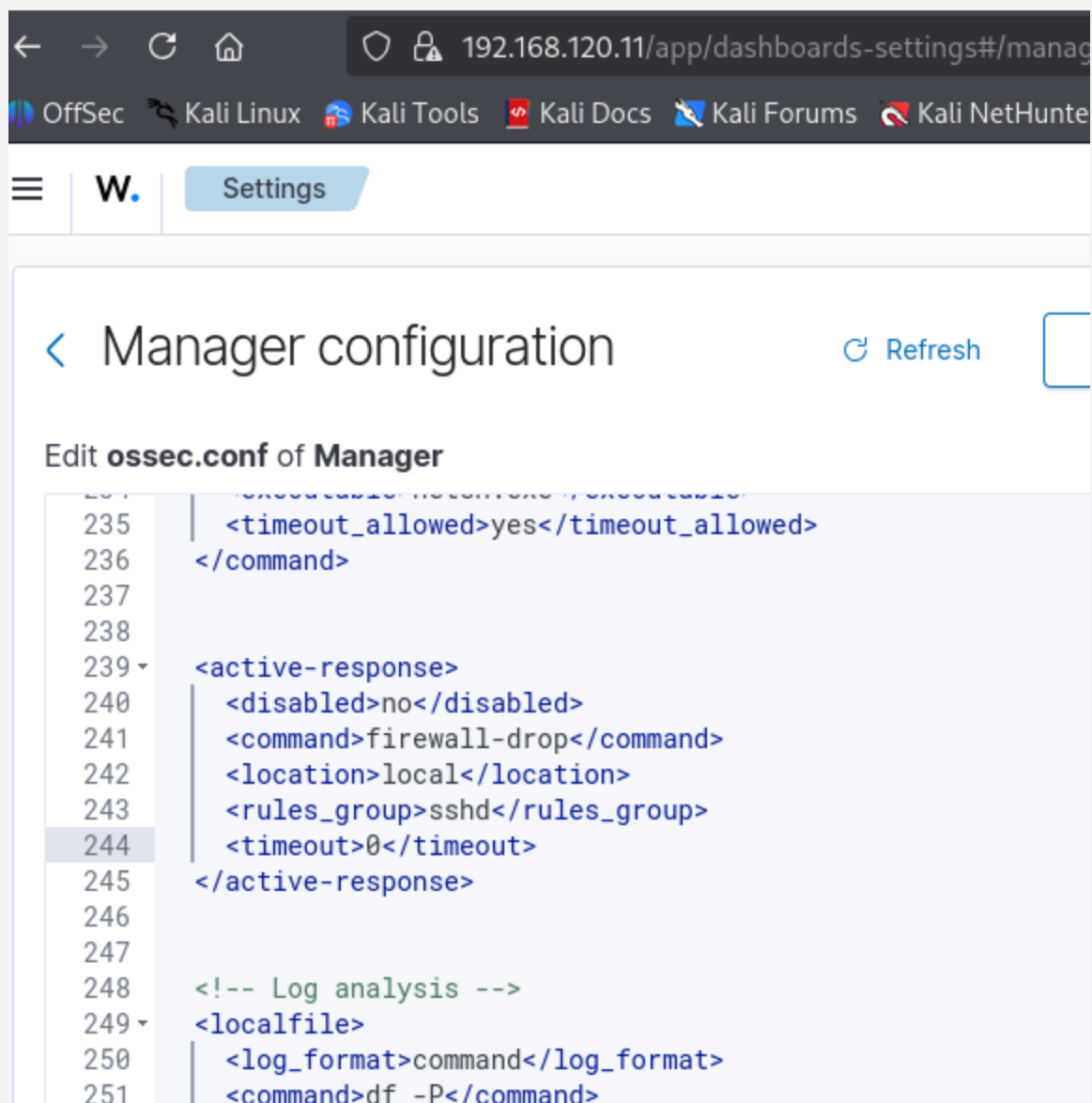
Exact commands used	
	nmap 192.168.120.0/24 nmap -sn 192.168.120.0/24 nmap -sS 192.168.120.1 nmap -sV 192.168.120.1 nmap -sU 192.168.120.1

tcp.port==22					
No.	Time	Source	Destination	Protocol	Length Info
381	25.287488783	192.168.128.10	192.168.130.2	TCP	74 50100 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2147751180 TSecr=0 WS=256
382	25.292633565	192.168.128.10	192.168.120.10	TCP	66 22 → 50060 [FIN, ACK] Seq=1804 Ack=1313 Win=64256 Len=0 TSval=1241753228 TSecr=2147751128
383	25.292696825	192.168.128.10	192.168.130.2	TCP	66 50060 → 22 [ACK] Seq=1313 Ack=1805 Win=66048 Len=0 TSval=2147751185 TSecr=1241753229
384	25.299579295	192.168.130.2	192.168.120.10	TCP	66 22 → 50046 [ACK] Seq=1804 Ack=1312 Win=64256 Len=0 TSval=1241753235 TSecr=2147751150
385	25.319800750	192.168.130.2	192.168.120.10	TCP	66 22 → 50086 [ACK] Seq=1 Ack=24 Win=65152 Len=0 TSval=1241753250 TSecr=2147751169
386	25.319890842	192.168.130.2	192.168.120.10	TCP	74 22 → 50100 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1241753258 TSecr=2147751180 WS=128
387	25.31992724	192.168.120.10	192.168.130.2	TCP	66 50100 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2147751212 TSecr=1241753258
388	25.327957322	192.168.120.10	192.168.130.2	SSHv2	89 Client: Protocol (SSH-2.0-libssh.0.11.3)
389	25.340119044	192.168.130.2	192.168.120.10	SSHv2	118 Server: Encrypted packet (len=52)
390	25.344163967	192.168.130.2	192.168.130.10	TCP	66 22 → 50046 [ACK] Seq=1804 Ack=1313 Win=64256 Len=0 TSval=1241753282 TSecr=2147751157
391	25.346388857	192.168.120.10	192.168.130.2	SSHv2	118 Client: Encrypted packet (len=52)
392	25.346774254	192.168.120.10	192.168.130.2	TCP	66 50074 → 22 [FIN, ACK] Seq=1312 Ack=1804 Win=66048 Len=0 TSval=2147751239 TSecr=1241753278
393	25.347595325	192.168.120.10	192.168.130.2	TCP	74 50100 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2147751240 TSecr=0 WS=256
394	25.489957820	192.168.120.10	192.168.130.2	TCP	66 [TCP Retransmission] 50074 → 22 [FIN, ACK] Seq=1312 Ack=1804 Win=66048 Len=0 TSval=2147751302 TSecr=1241753278
395	25.431373015	192.168.130.2	192.168.120.10	TCP	66 22 → 50100 [ACK] Seq=1 Ack=24 Win=65152 Len=0 TSval=1241753290 TSecr=2147751220
396	25.434783769	192.168.130.2	192.168.120.10	SSHv2	550 Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=284)
397	25.434785161	192.168.130.2	192.168.120.10	TCP	66 22 → 50046 [FIN, ACK] Seq=1804 Ack=1313 Win=64256 Len=0 TSval=1241753306 TSecr=2147751157
398	25.436577922	192.168.120.10	192.168.130.2	TCP	66 50946 → 22 [ACK] Seq=1313 Ack=1805 Win=66048 Len=0 TSval=2147751329 TSecr=1241753306
399	25.440215100	192.168.120.10	192.168.130.2	SSHv2	82 Client: New Keys
400	25.444565518	192.168.130.2	192.168.120.10	TCP	66 22 → 50074 [ACK] Seq=1804 Ack=1312 Win=64256 Len=0 TSval=1241753378 TSecr=2147751239
401	25.448859820	192.168.130.2	192.168.120.10	TCP	74 22 → 50100 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=1241753381 TSecr=2147751240 WS=128
402	25.448980155	192.168.120.10	192.168.130.2	TCP	66 50100 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2147751341 TSecr=1241753381
403	25.449348802	192.168.120.10	192.168.130.2	SSHv2	80 Client: Protocol (SSH-2.0-libssh.0.11.3)

Filter applied	tcp.port == 22 The filter was used to display only traffic associated with TCP port 22, which is the default port used by the Secure Shell (SSH) protocol. This allowed the investigation to focus specifically on SSH connection attempts and authentication-related activity while removing unrelated network traffic.
Time	The time range was approximately between 25.287 seconds – 25.319 seconds

Appendix C





```
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-15 15:45:46

(kali@kali)-[~]
$ hydra -L users.txt -P password.txt -t4 -V ssh://192.168.130.11
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret s
ervice organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethi
cs anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-15 15:45:58
[DATA] max 4 tasks per 1 server, overall 4 tasks, 16 login tries (l:4/p:4), ~4 tries per task
[DATA] attacking ssh://192.168.130.11:22/
[ERROR] could not connect to ssh://192.168.130.11:22 - Timeout connecting to 192.168.130.11
```

Key analytical questions

Network Design

1. Which assets belong in the LAN, WAN, and DMZ zones?

The WAN is the external network, primarily the Internet, from which remote users, telehealth patients, and external services connect to the organisation. The patient portal, web server, and any publicly available services belong within the DMZ and LAN contains the organisation's trusted internal assets, including clinical workstations, administrative computers, Electronic Health Record (EHR) servers, database servers, domain controllers, file servers, and the Wazuh Manager.

Firewall Security

2. Which traffic should be allowed between zones?

Communication among network zones must adhere to the least privilege principle. Internet users should be allowed to access only services located within the DMZ, like the patient portal via HTTPS (TCP port 443). The DMZ must interact with internal systems only when necessary such as obtain patient data from the EHR database using strictly managed firewall regulations. Authorised personnel within the LAN should have access to business applications, file servers, printers, and clinical resources necessary to carry out their responsibilities. SSH administrative access must be limited to authorised management devices and reliable administrators.

3. Which traffic should be blocked?

All communication that is not explicitly necessary for business activities should be denied. Internet access to internal LAN systems, such as the EHR database, domain controllers, and employee workstations, must never be allowed. Likewise, the DMZ must not have unlimited access to the internal network. Unnecessary services, inactive ports, peer-to-peer connections, and unauthorised remote access must be restricted as well. Limiting lateral movement among internal network segments aids in stopping attackers from accessing essential healthcare systems if a single endpoint is compromised.

Threat Detection

4. How can suspicious internal traffic be identified?

By integrating network monitoring, endpoint logging, and correlation of security events, it is possible to identify dubious internal behavior. In this initiative, Wazuh

gathered authentication logs, system events, and security warnings from monitored devices, while Wireshark recorded network traffic for in-depth examinations. pfSense firewall logs additionally offered insights into connection attempts across network segments. Signs of questionable activity comprise multiple unsuccessful login attempts, unanticipated interactions with sensitive servers, atypical port scanning, excessive outgoing traffic, misuse of privileged accounts, and connections happening beyond regular working hours. Setting a standard of normal network behavior allows analysts to recognise abnormalities that could signal harmful actions.

Risk Assessment

5. What business risks exist if segmentation is not implemented?

In the absence of proper network segmentation, a breach impacting one workstation could quickly propagate across the entire organisation. Threat actors could navigate to clinical systems, gain access to Electronic Health Records (EHRs), compromise medical devices, or disrupt vital healthcare services. These events might lead to the exposure of protected health information (PHI), financial losses, regulatory fines for HIPAA violations, harm to reputation, and diminished patient trust. Operational interruptions from ransomware attacks or system failures may also postpone clinical treatment and adversely impact patient safety.

Security Improvement

6. What additional controls should be introduced to strengthen security?

While network segmentation greatly enhances security, implementing extra controls could offer even better protection. Multi-factor authentication must be applied for privileged accounts and remote access services. Firewall rules must be evaluated consistently to confirm they stay aligned with organisational needs, and Secure Shell (SSH) access ought to be strengthened using key-based authentication and limited to trusted management hosts. The organisation must implement a systematic vulnerability management program that encompasses frequent patching, vulnerability assessment, and penetration testing. Endpoint Detection and Response (EDR) solutions would enhance endpoint visibility, whereas continuous cybersecurity awareness training would lower the chances of phishing attacks and various types of human error. Ultimately, keeping an incident response plan with well-defined protocols would enhance the organisation's preparedness to address upcoming security incidents.

SOC Perspective

7. How would a SOC analyst investigate an alert generated from an internal workstation?

A SOC analyst would start by examining the alert produced in the SIEM to understand its seriousness, the rule that triggered it, the impacted host, and the related user account. The analyst would subsequently link the alert with other evidence sources, such as Windows or Linux event logs, firewall logs, Wazuh alerts, and Wireshark packet captures, to create a timeline of the activity. The following step involves assessing whether the behaviour was valid or harmful by analysing processes active on the workstation, login attempts, network connections, and recent system modifications. Should any malicious activity be verified, the analyst would separate the compromised device from the network to avert additional breaches, safeguard evidence for forensic examination, and initiate containment measures. After elimination and restoration of the system, the incident will be documented, insights gained will be noted, and suggestions will be applied to avoid comparable occurrences in the future. This structured investigation procedure reflects typical Security Operations Centre (SOC) methods and enhances competent incident response ability in a healthcare setting.